

From Spectrum to Colour

Internal documentation · 2026-08-24 · measured on runs 001 and 002 , then tested against the 88 labelled archive runs (§15)

How a measured spectrum becomes the colour chips the app shows, what each chip can and cannot tell you, and why the set looks the way it does. **No colour-science background is assumed** — §§1–6 build the geometry from the beginning, and §0 fixes the notation the rest of the document uses.

★ **The one result everything else follows from.** The chips that are *dilution-invariant* are, for exactly that reason, blind to what distinguishes two pumpkin oils: invariance and discriminating power are the same property seen from two sides. §3.1 measures the consequence — the absorbed chromaticity points the same way, $244.06 \pm 1.25^\circ$, on every oil in the archive. That is chemistry, not software, and no chip can be built around it.

⇒ What the colour chips are *for* is therefore a visual aid, and the one that does that job — the transmitted colour at a real viewing path — leads the set. The metrics that do discriminate live in

`SPEC_metric_research.md` .

Map. §§1–7 the geometry, from the chromaticity diagram to the chip that separates the oils · §8 Beer–Lambert, and what the reference does and does not fix · §9 the five-stage correction chain · §10 the family tree · §§11–12 the design in use · §13 the implementation record · §§14–15 glossary and provenance.

△ Two naming conventions run through the text. The phases **P0–P8** are the implementation steps, defined in §13. Chips are named as they are **today** everywhere except §11.4's mapping table, which gives the older names, and the note in §13.4 about reloaded saved runs.

CONTENTS

0. Notation

1. Colour is a place on a map, not a number

2. Direction and distance: hue and purity

- 2.1 Neither purity nor dominant wavelength separates these oils
⇒ What §2.1 leaves standing

3. What "dilution-invariant" costs

- 3.1 And the direction is a constant of the pigment, measured

4. The complement, and why it cannot help

- 4.1 The complement routinely leaves the diagram
- 4.2 But the hue survives all of it — the "4°" claim, re-measured on 88 runs
- 4.3 Why it still cannot separate the oils

5. The gamut wall: where 1.33° becomes 2.68°

6. LCh — the model the $\times 3$ path chip reports in

7. The chip that *does* separate them

8. Beer–Lambert: what the law does and does not say

- 8.1 The identity
- 8.2 Where the physics — and the convention — enter
- 8.3 For a colour chip, the reference is the illuminant
- 8.4 "3 cm" is a claim the vessel does not support

9. How a chip is corrected — the five stages

- 9.1 Stage 3, the one that mattered
- 9.2 For a year, stage 3 reached ONE chip out of eleven

10. The family tree

11. Why the chip set looks the way it does

- 11.1 It does not report a dominant wavelength
- 11.2 Chips whose colour cannot be shown are marked
- 11.3 The processing rungs live in a sub-tab
- 11.4 The names say what each chip is the colour OF
- 11.5 Every chip shares one CIE path
- 11.6 None of this makes the invariant chips separate the oils

12. The colour model in use: Lab, with θ_w as the hue

- 12.1 What it forces
- 12.2 Order of operations — and it was the right order
- 12.3 Measured: on the invariant chips, LCh costs hue accuracy

13. AS BUILT — 2026-08-24

- 13.1 The chip set as it now reads
- 13.2 What P1 moved, and what it did not
- 13.3 The whole archive regenerated — and the regression it caught
- 13.4 Owed

14. Glossary

15. Provenance

Appendix A — Why the SOLVENT changes what the EYE can see

- A.1 The instrument cannot see what the eye sees — by construction
- A.2 The dominant mechanism: refractive-index matching
- A.3 Why scattering destroys a colour DIFFERENCE, not just brightness
- A.4 The second mechanism, which also favours sunflower



A.5 Observation 2 — a mechanism proposed, tested, and refuted the same day

△ A.5.1 What the test could not do

A.6 △ Where to hold back, and the experiments

0. Notation

Symbols are used consistently throughout, with the subscript naming what the quantity is measured *against*.

symbol	meaning
$S(\lambda), R(\lambda)$	the sample and reference captures
$T(\lambda) = S/R$	transmittance — the fraction of the reference beam that gets through
$A(\lambda) = -\log_{10} T$	absorbance
$\varepsilon(\lambda), c, \ell$	molar absorptivity, concentration, optical path
k	the path multiplier — $k = 3$ on the <code>×3 path</code> chip
X, Y, Z	CIE tristimulus values; Y is luminance
(x, y)	chromaticity — XYZ with luminance divided out
(x_w, y_w)	the white point ; D65 2°, $(0.313, 0.329)$
θ_w	the direction of a chromaticity from the white point
r_w	its distance from the white point
R_w	the distance from white to the spectrum locus along that same ray
λ_d	dominant wavelength — where the ray meets the locus
$p_e = r_w / R_w$	excitation purity
p_c	<i>colorimetric</i> purity — a different, luminance-weighted definition
L^*, C_{ab}^*, h_{ab}	CIE Lab lightness, chroma and hue angle

△ θ_w and h_{ab} are **both** hue angles and they are **not the same quantity**: θ_w lives on the chromaticity diagram, h_{ab} in Lab. §12 turns on the difference.

1. Colour is a place on a map, not a number

The eye has three cone types. Any light that enters it, however complicated its spectrum, is reduced by those three cones to **three numbers** — in the standard formalism `X`, `Y`, `Z`. That is the whole of colour vision: an infinite-dimensional spectrum collapsed onto three numbers. Two completely different spectra with the same three numbers look identical.

Of those three, `Y` is **luminance** — how bright. The other two carry *which colour*. Divide them out,

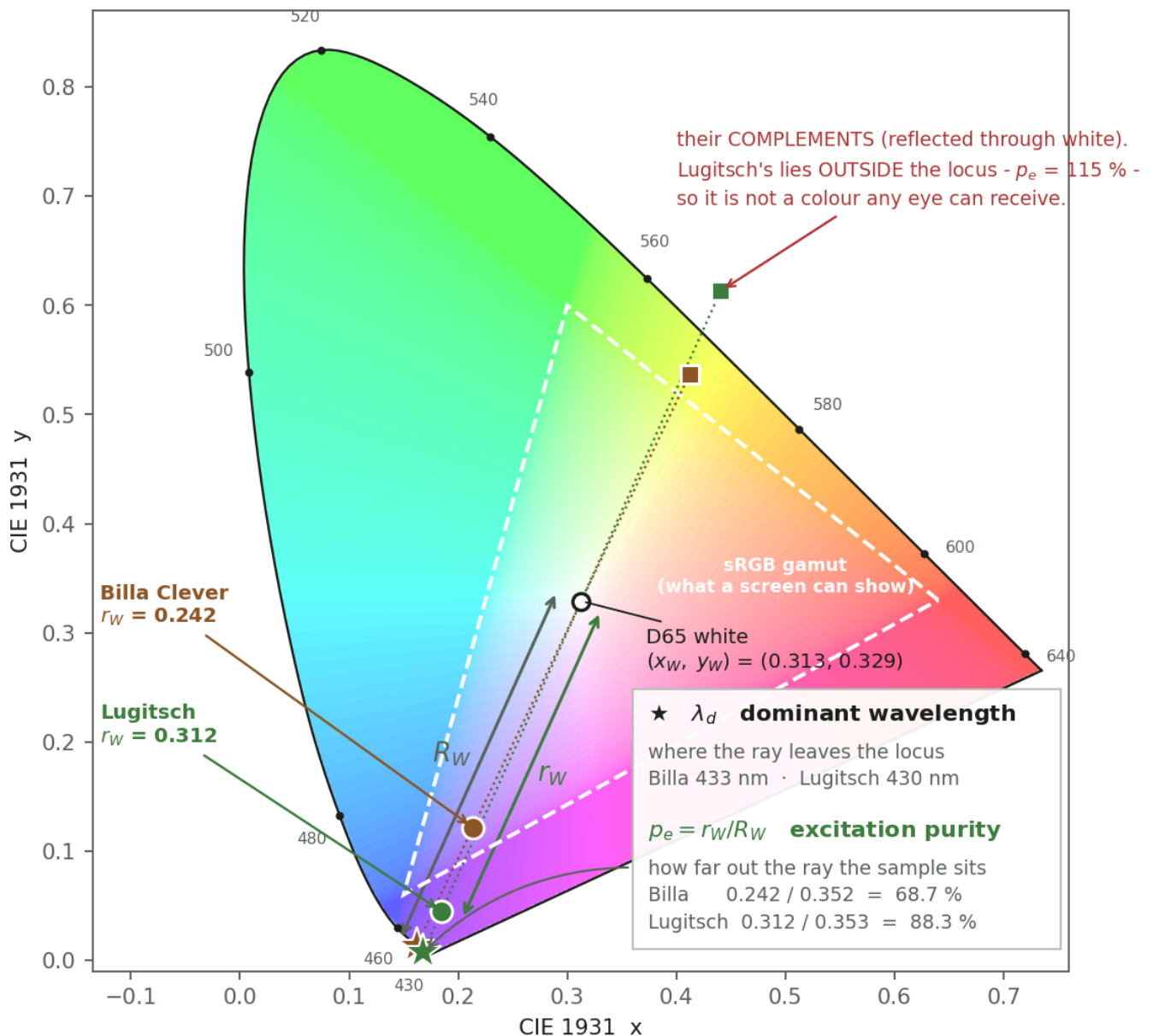
$$x = \frac{X}{X+Y+Z} \quad y = \frac{Y}{X+Y+Z}$$

and you are left with **chromaticity**: a point (x, y) on a two-dimensional map. This is the famous horseshoe. Its curved edge is the **spectrum locus** — the pure single wavelengths, 430 nm at the bottom left round to 640 nm at the right. Its straight bottom edge is the purples, which have no wavelength of their own. Everything a human can see lies inside.

★ **The single most important thing about this map:** dividing by $X+Y+Z$ **deleted the brightness**. A dark brown and a pale beige of the same hue land on the *same point*. Remember that — it is the answer to question 3.

Near the middle sits **white**. We use the D65 white point, $(0.313, 0.329)$ — roughly noon daylight.

Both oils sit on the SAME ray out of white - only the DISTANCE differs



The two absorbed chromaticities differ by 0.30° of θ_w and by 36 % of r_w . Every chromaticity chip reports the direction and discards the distance.

Figure 1 — the CIE 1931 chromaticity diagram, with both oils, their complements, and the λ_d / p_e construction drawn on the ray.

Both oils' absorbed colours are on that map. And they sit **on the same line out of white**.

2. Direction and distance: hue and purity

A point on a map can be given as x and y , or as **a direction and a distance** from some origin. Take white as the origin, and the two polar coordinates get names:

- The **direction** θ_W — the compass bearing from white — is essentially **hue**. Follow it outward until it meets the spectrum locus and you read off the **dominant wavelength** λ_d : the single pure colour this sample most resembles.
- The **distance** r_W from white, taken as a fraction of the distance R_W to the locus on that same ray, is the **excitation purity** $p_e = r_W/R_W$: 0 % at white, 100 % on the locus. It is how *strongly* coloured the sample is, independently of how bright it is.

△ Both depend on the chosen white point, not on the sample alone: Billa reads $\lambda_d = 444$ nm and $p_e = 65.7$ % under D65, but **451 nm and 69.6 % under D50**. And "purity" is ambiguous — *colorimetric* purity p_c is luminance-weighted and returns **7.2 %** on the very same chromaticity.

Measured on the two runs:

	Billa Clever 001	Lugitsch 002	difference
absorbed chromaticity x	0.21353	0.18484	
absorbed chromaticity y	0.12164	0.04485	
direction θ_W	244.44°	245.77°	1.33°
dominant wavelength λ_d	444 nm	432 nm	12 nm
distance r_W	0.2299	0.3116	36 %
purity p_e	65.7 %	88.4 %	23 points

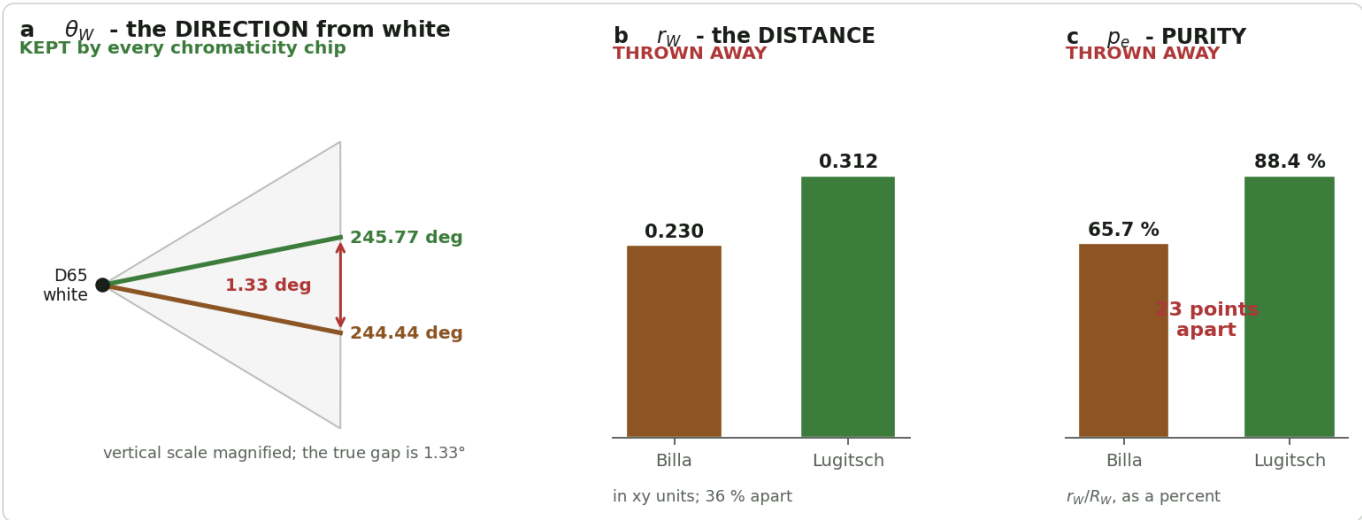


Figure 2 — the same two points in polar form: θ_w is kept by every invariant chip, r_w and p_e are thrown away.

Read the table again. The two oils differ by **one and a third degrees** of direction and by **23 points of purity**. They are two points on one ray, at different distances.

Why they share a direction is physics, not coincidence. Both oils carry the same pigment family — chlorophyll derivatives. Both absorbance curves are dominated by the same enormous Soret band at 420–450 nm. A colour is a weighted integral over the whole spectrum, and when one feature dominates both curves, both integrals point the same way. What differs between the oils is *how much of the absorbance sits outside that band* — and that is a distance from white, not a direction.

2.1 Neither purity nor dominant wavelength separates these oils

The table above is **two runs**, and the 23-point purity gap in it does not survive contact with more data. Two findings follow, in the order they were established — the second retracts the first.

First: across all seven archived fills of these two oils, purity does not separate them.

oil	fills	λ_d	p_e
Billa Clever	3	445.7 ± 2.9 nm — [444, 449]	63.4 ± 5.3 % — [57.4, 67.2]
Lugitsch	4	436.8 ± 3.3 nm — [432, 439]	76.0 ± 8.4 % — [69.8 , 88.4]

The p_e ranges **nearly touch**, and the 88.4 % in §2's table is Lugitsch's *outlier* — the highest of its four fills. λ_d , by contrast, looked clean on those seven: a **5 nm gap** with no overlap — which made it the obvious quantity to lead with instead.

Second: that is wrong too, and the archive says so. Seven fills of one oil pair, one solvent, three days, is not evidence. Re-run over the **88 labelled isopropanol runs** of the report archive (diagnostics/dominant_wavelength_archive.py, same labelling as SPEC_metric_research.md §12):

quantity	green (n=55)	brown (n=33)	Cohen's d	
λ_d — the pre-2026-08-24 align path	35.1 ± 477.0 nm	231.6 ± 382.3 nm	−0.44	meaningless, see below
λ_d — as shipped now	444.5 ± 9.2 nm	445.2 ± 7.0 nm	−0.08	OVERLAP
p_e — the pre-2026-08-24 align path	72.7 ± 5.1 %	69.7 ± 5.7 %	0.56	OVERLAP
p_e — as shipped now	61.0 ± 10.9 %	57.0 ± 11.4 %	0.36	OVERLAP

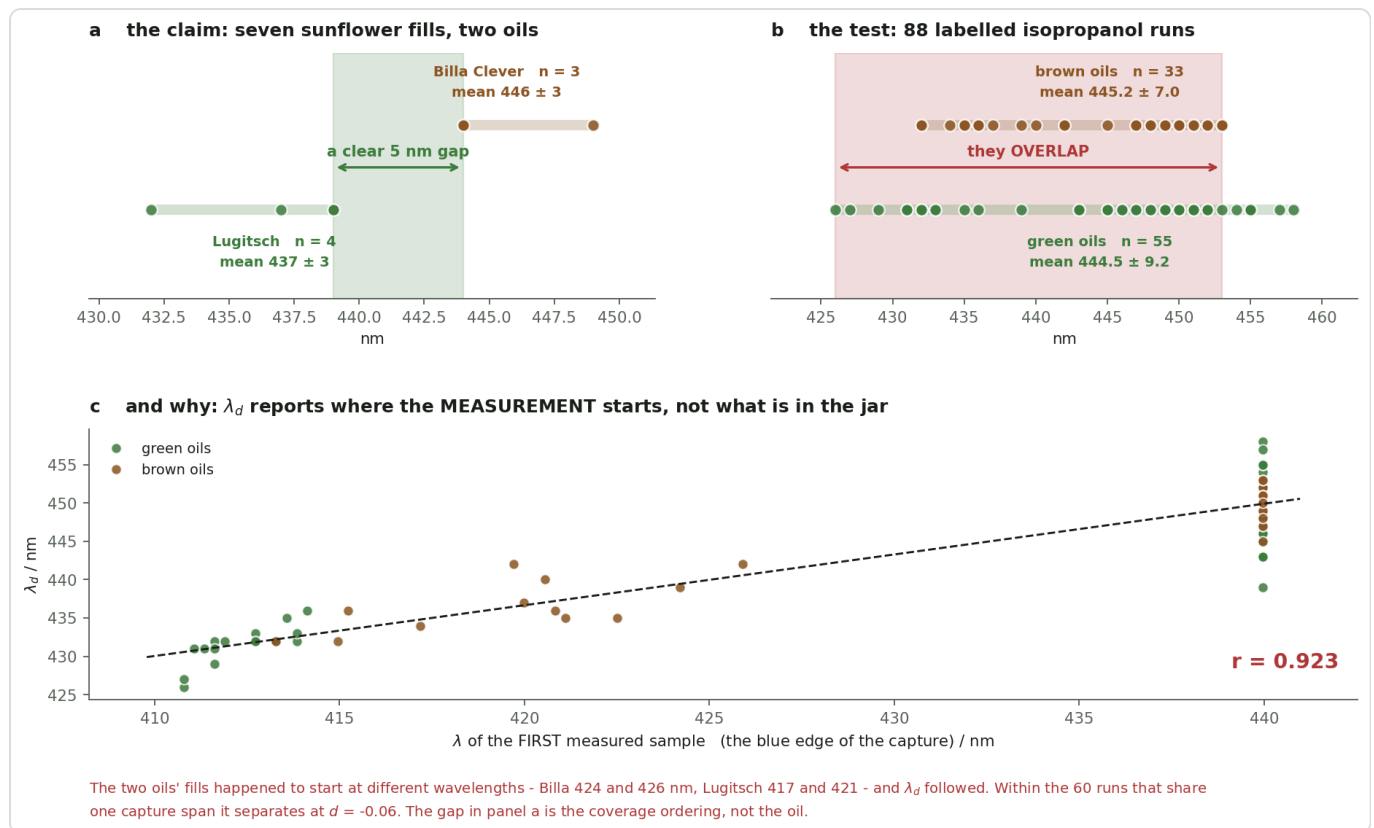


Figure 3 — the claim, the test, and the confound that produced it.

λ_d fails three ways over:

1. **It does not exist on 31 % of the archive.** 27 of the 88 runs return a *negative* value — colour's convention for "the ray exits through the purple line, so there is no dominant wavelength" — across nine different series. A quantity that is undefined on a third of the corpus cannot be a reported number. (The shipped padding makes it defined everywhere: 0 of 88.)
2. **Where it is defined, it does not separate.** $d = -0.08$.
3. **It is 92 % an artefact of the capture.** $r(\lambda_d, \lambda_{\text{first}}) = 0.923$ over 88 runs. The clamped runs (first sample at 440 nm, $n = 60$) read 449.9 ± 3.6 nm; the wide-blue runs (411–426 nm, $n = 28$) read 433.7 ± 3.9 nm. λ_d reports where the measurement starts, not what is in the jar.

⇒ And that is exactly what produced the gap. Billa's fills happened to begin ~5 nm later than Lugitsch's — Billa at 424 and 426 nm, Lugitsch at 417 and 421 — and λ_d followed. **Within the 60 runs that share one capture span, $d = -0.06$.** The separation was the coverage ordering.

△ Why this is physically unsurprising: λ_d is set by the **Soret**, the sharpest feature in the spectrum and the one nearest the blue edge. Truncating the blue end truncates exactly the feature that fixes the ray's direction.

★ **One hypothesis died cleanly and usefully:** the **RELATIVE ceiling** was the other suspected confound — it clips the Soret when it fires, and λ_d would move more than anything else in the chip set. It fired on **0 of 88 runs**. Disposed of, not deferred.

⇒ What §2.1 leaves standing

- **Neither half of (λ_d, p_e) discriminates on the archive**, so the app describes a chromaticity with them but claims nothing from them (§11.1).

- The **Absorbed · purity** row is a visual aid, not a discriminator — $d = 0.55$, overlapping, on 88 runs. It is kept because it is the honest replacement for $S \equiv 100$, and because it predicts exactly when the complement stops being a colour (§4.1).
- ★ The whole chip family is now measured, not merely suspected, to be non-discriminating, which is §1c of `SPEC_color_retrieval.md` with numbers behind it. That *strengthens* the case for putting `Transmitted from absorbance · ×3 path first` (§11.4): the one chip whose job is to look like the oil is also the only one carrying anything.
- ★ Checked, in §4.1: 27 runs whose ray exits the purple line suggested `Absorbed-complement` might be an *imaginary* colour on a large slice of the archive. It is: outside the locus on 10 % of runs, at or past the edge of vision on 25 %.

3. What "dilution-invariant" costs

The intrinsic chips exist to answer: *what colour is this oil, regardless of how much we diluted it?* The mechanism is exactly the one in section 1 — **drop the luminance** and keep only chromaticity. Double the concentration and `X`, `Y`, `Z` all scale together; `x` and `y` do not move. Invariance achieved.

But the chips go one step further. The `· hue-norm` variants re-render the colour at a **fixed saturation and lightness**, so that only the hue can move between chips. That throws away the *distance* as well.

⊖ So the eight `Absorbed` and `Absorbed-complement` chips report the direction and discard the distance — and on these two oils, the distance is where the entire difference lives. The chip is a compass. You asked it which of two towns is bigger.

This is not a defect to be patched. It is the price of the invariance, and it was priced correctly: a chip that cannot be fooled by dilution cannot be moved by concentration either.

★★ 3.1 And the direction is a constant of the pigment, measured

θ_W over the **88 labelled archive runs** — two oil classes, a year of fills, a mechanical rebuild, several dilutions and two rigs:

$$\theta_W = 244.06 \pm 1.25^\circ \quad [241.33 \dots 246.19]$$

★ **Five degrees of total spread.** Every pumpkin oil absorbance measured on this instrument points the same way out of white, to within about a degree. Green oils $244.06 \pm 1.36^\circ$, brown $244.08 \pm 1.06^\circ$ — Cohen's $d = -0.02$, overlapping. Within the 60 runs that share one capture span: 243.33 ± 0.88 against 243.35 ± 0.45 . **Indistinguishable to two decimal places.**

⇒ **This is the physical reason the invariant chips cannot work, and it is not a software problem.** The direction is fixed by the *chemistry* — it is a constant of the chlorophyll-derivative family, set by the Soret band that dominates every one of these spectra. What differs between one oil and another is the **radius**: how strongly, and how purely, that same absorbance sits. And the radius is precisely what a dilution-invariant chip is built to discard.

△ So the eight hue-norm chips are not merely redundant with each other (§2). They are reporting a quantity that is **very nearly a constant of the pigment class**, and printing it to two decimals. ⇒ §11.3: they left the

headline list for a sub-tab.

4. The complement, and why it cannot help

Absorbed-complement is not the absorbed colour — it is its **complement**: the other half of the light. If the oil absorbs blue-violet, what passes through is the yellow-green that is left over.

We compute it as a **reflection through the white point**:

$$(x, y) \rightarrow (2x_W - x, \quad 2y_W - y)$$

which is the colorimetric "mixing-to-white" opposite: the chromaticity that, mixed with the original in equal measure, gives white. The earlier implementation flipped the HSL hue by 180° instead; §4.2 measures both against the transmitted colour they stand in for.

4.1 The complement routinely leaves the diagram

Reflecting through white preserves the radius, and nothing constrains the reflected point to stay inside the horseshoe. Measured over the 88 labelled archive runs (diagnostics/dominant_wavelength_archive.py):

where the complement lands	runs	
below 90 % purity — comfortably inside	66	75 %
90–100 % — inside, but at the edge of human vision	13	15 %
over 100 % — outside the spectrum locus	9	10 %

Mean complement purity is 77.9 ± 14.6 %, worst 107.7 %. So on **25 % of the archive the Absorbed-complement chip sits at or beyond the boundary of what an eye can receive** — and where it is beyond it, C^*_{ab} and h_{ab} describe an imaginary stimulus. (Lugitsch's 115.3 % on the sunflower fill is worse than anything in the isopropanol corpus, which is consistent: it also has the highest absorbed purity.)

★ **The failure is exactly predictable, and free to detect.** Absorbed and complement purity correlate at $r = 1.000$ — the complement's radius *is* the absorbed radius, only measured against a shorter locus distance on the opposite ray. The two populations do not overlap:

	absorbed p_e
complement real (n = 79)	57.2 ± 9.4 % — [35.1, 75.3]
complement imaginary (n = 9)	79.5 ± 2.2 % — [76.9 , 82.5]

⇒ **absorbed p_e above ≈ 76 % ⇒ the complement is not a colour.** The predictor is a number the chip set already computes, so the §11.2 marker costs nothing here.

△ **And the failure is asymmetric between the two classes the chip exists to compare** — green oils 79.8 ± 14.2 % complement purity and 13 % imaginary, brown 74.6 ± 14.8 % and 6 %. A chip that degrades more for one class than the other is a poor instrument for telling them apart.

★★ 4.2 But the hue survives all of it — the "4°" claim, re-measured on 88 runs

`SPEC_capability_proof.md` option (b) chose the white-point complement over the `+180°` HSL flip because it lands "*~4° from the true perceived hue, versus ~34°*" — measured on **four** runs (K/L/M/N). Given §4.1, that number needed re-testing where the complement is worst. Reference = the transmitted chromaticity, derived as $T = 10^{-A}$ from the same trace (exact to 6×10^{-17}) so both sides share one wavelength span:

hue coordinate	error vs the transmitted colour	
★ θ_W — the chromaticity direction	$2.50 \pm 1.90^\circ$ — [0.24, 8.68]	
H_{hsl} — the retired HSL readout	$4.15 \pm 3.30^\circ$ — [0.01, 14.80]	
h_{ab} — CIE Lab	$5.60 \pm 4.00^\circ$ — [0.13, 15.15]	see §12.3
the rejected <code>+180°</code> flip	$52.69 \pm 4.91^\circ$ — [44.59, 62.03]	

★ "*~4°*" was right — better than right: **2.50° over 88 runs**. And the `+180°` flip is far worse than the 34° it was charged with: **52.7°** . Option (b) was the right call by a wide margin.

★★ And it does not degrade where the complement is imaginary — $1.68 \pm 0.36^\circ$ there, against $2.60 \pm 1.98^\circ$ where the complement is real. Slightly *better*. Which is exactly what the geometry predicts: a point reflection preserves direction whatever the radius, so leaving the locus costs renderability, not angle.

⇒ **The first positive result for the invariant chips in this document.** `Absorbed-complement` is a good hue estimator even when it is not a colour. **Its hue is trustworthy; its chroma and its swatch are not.** Report θ_W , suppress or mark the rest.

4.3 Why it still cannot separate the oils

★ A point reflection preserves the direction and the distance. It maps a ray to the exactly opposite ray and keeps the radius. Measured: the complements come out at **64.44°** and **65.77°** — the same 1.33° gap, mirrored, and the same radii 0.2299 and 0.3116.

The complement therefore cannot create an angular difference that was not already there. That is half of the answer to question 4.

5. The gamut wall: where 1.33° becomes 2.68°

The other half is that these colours **cannot be shown on a screen at all**.

A monitor makes colour by mixing three fixed primaries. The set of chromaticities it can reach is the dashed triangle in figure 1 — the **sRGB gamut**. Everything outside it is real, visible, and unreachable. Both oils' absorbed chromaticities are far outside; so are both complements.

To draw a swatch anyway, the code converts to R, G, B and **clamps each channel into the range 0 to 1**. That is like pushing a point onto the nearest wall of a box. Two points far apart *outside* the box can land close together *on* it.

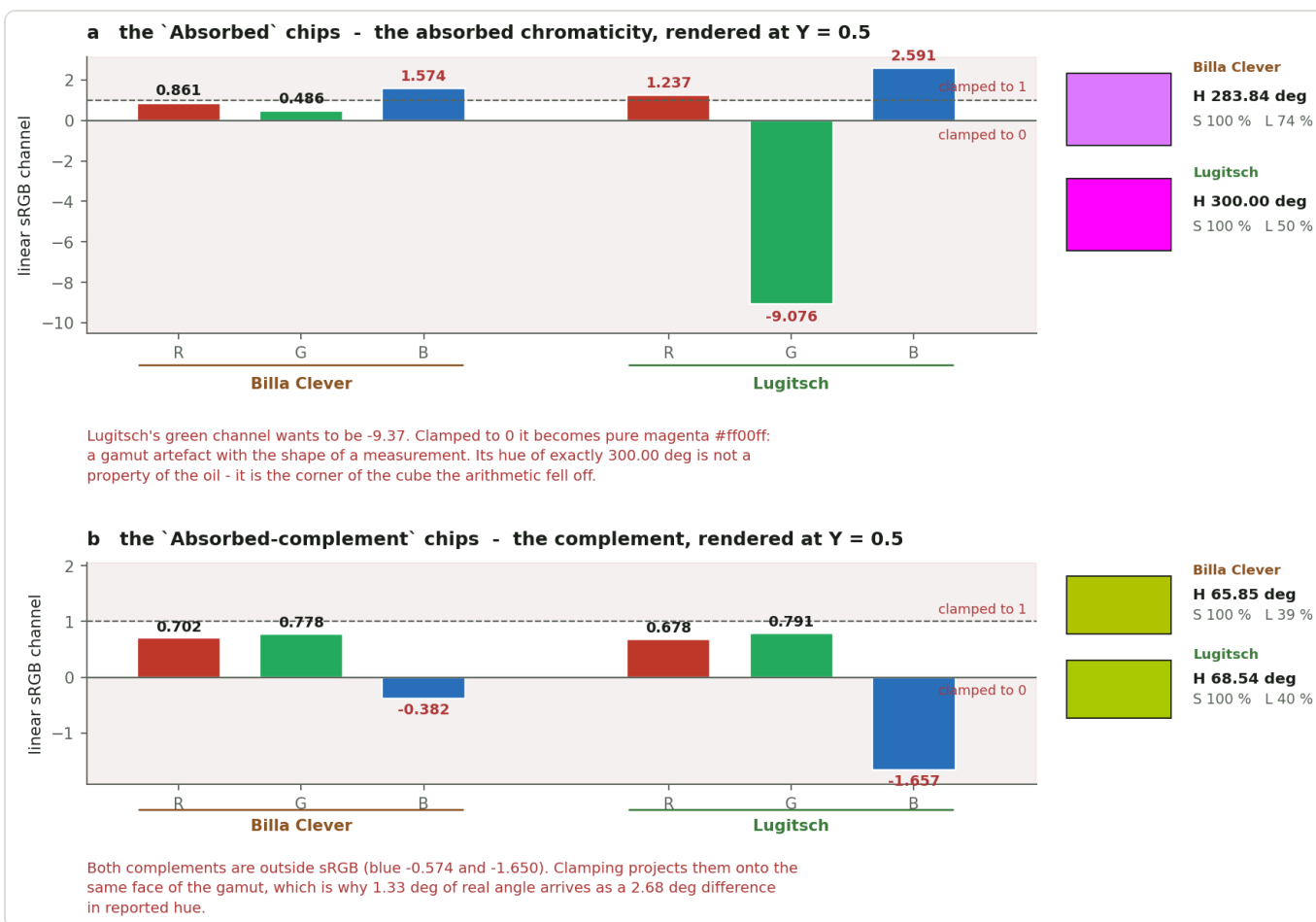


Figure 4 — the linear sRGB channels before clamping. Everything shaded red is outside what a screen can produce.

Look at the numbers before the clamp:

chip	R	G	B	after clamping
Billa Absorbed	0.861	0.486	1.574	H 283.84°
Lugitsch Absorbed	1.237	-9.076	2.591	H 300.00°
Billa Absorbed-complement	0.702	0.778	-0.382	H 65.85°
Lugitsch Absorbed-complement	0.678	0.791	-1.657	H 68.54°

⊖ **Lugitsch's green channel wants to be -9.076.** Clamped to zero, its chip comes out as H 300.00° · S 100 % · L 50 % — exactly pure magenta, #ff00ff . That is not a property of the oil. It is the corner of the RGB cube the arithmetic fell off. The apparent 283.8° vs 300.0° difference on the Absorbed row is a rendering artefact, not a measurement.

And for the complements: two chromaticities 1.33° apart, both projected onto the same face of the gamut, arrive as **2.68° of reported hue**. Question 4, answered:

1.33° of real difference, compressed by clamping into 2.68° of reported hue. The number is arithmetically correct and informationally empty.

⚠ **S ≡ 100 % has the same cause.** The gamut boundary is precisely where $\min(R, G, B) = 0$, which is HSL saturation 100. Every real oil read $S = 100 \%$. That is why Absorbed · purity exists: purity is

defined on the chromaticity map, which has no gamut, so it returns a real number where `S` returns a wall.

6. LCh — the model the x3 path chip reports in

L*a*b* is CIE **L*a*b*** written in polar coordinates. Same colour, different notation. It is the model the x3 path chip reports in, and §12 describes how Lab came to render every chip.

L*a*b* was designed so that **equal numerical steps look like equal perceptual steps** — unlike X, Y, Z (physics) or HSL (a 1970s convenience for painting on displays). Its three axes are lightness **L***, a green-to-red axis **a***, and a blue-to-yellow axis **b***. Because "how much **a*** and how much **b***" is awkward to read, the pair is usually given as a length and an angle instead:

symbol	name	range	reading
L^*	lightness	0 ... 100	0 = black, 100 = diffuse white
C^*_{ab}	chroma = $\sqrt{a^{*2} + b^{*2}}$	0 ... ~130	0 = neutral grey; higher = more colourful
h_{ab}	hue angle = $\arg(a^* + i b^*)$	0 ... 360°	0° red, 90° yellow, 180° green, 270° blue

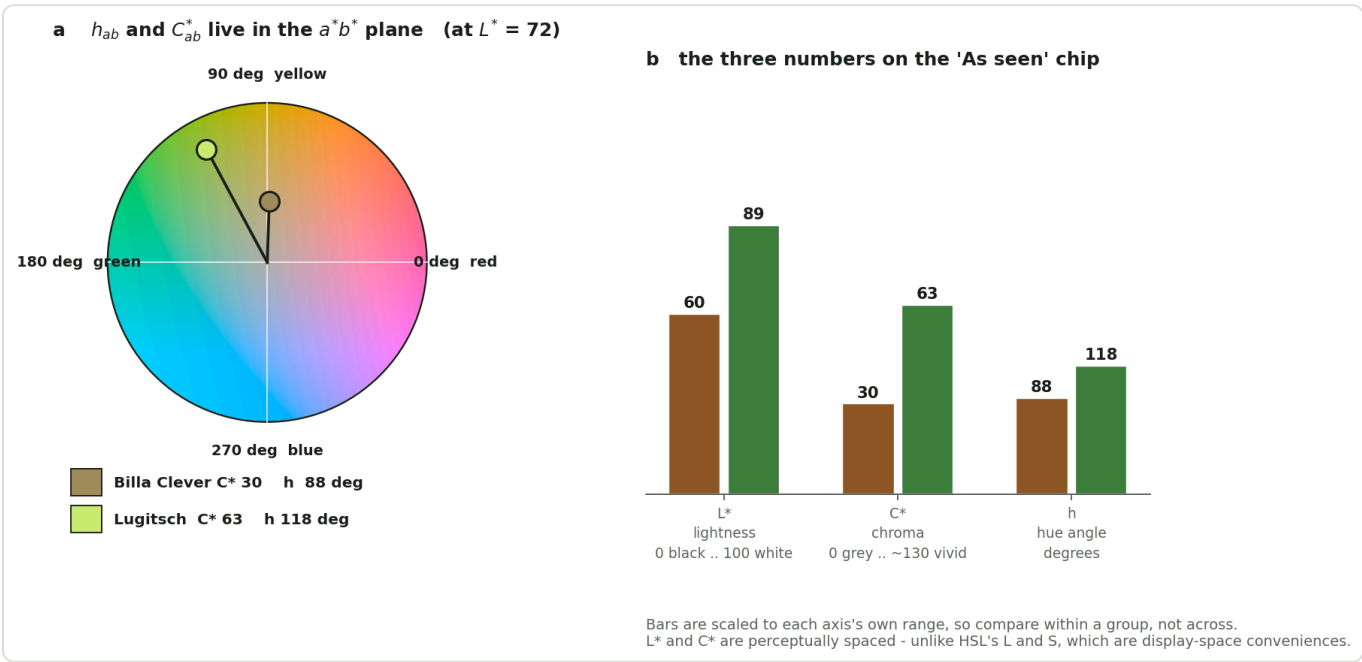


Figure 5 — $L^*C^*_{ab}h_{ab}$: the perceptually spaced model, and the three numbers it reports.

So $L^* 60 \cdot C^* 30 \cdot h 88^\circ$ reads as *a medium-dark, moderately colourful yellow* — khaki brown. And $L^* 89 \cdot C^* 63 \cdot h 118^\circ$ is *light, strongly colourful, yellow-green*.

★ **C*** is the honest replacement for HSL's **S**. It has no gamut, so it cannot pin at 100. **L*** is the honest replacement for HSL's **L**. This is the model to quote whenever a number leaves the app.

7. The chip that *does* separate them

Transmitted from absorbance · x3 path breaks the invariance on purpose, and that is the whole trick. Instead of dropping luminance, it renders the absorbance at a **declared viewing path** of 3 cm — a 4 ml eprouvette seen end-on, the geometry in which the two oils are obviously different to the naked eye. Beer—

Lambert says the transmitted fraction at path `k` is

$$T(\lambda) = 10^{-k \cdot A(\lambda)}$$

and `k = 3` is applied, then luminance is **kept**. §8 takes that formula apart — what it assumes, what it does not, and why the chip is named `• ×3 path` rather than `• 3 cm`.

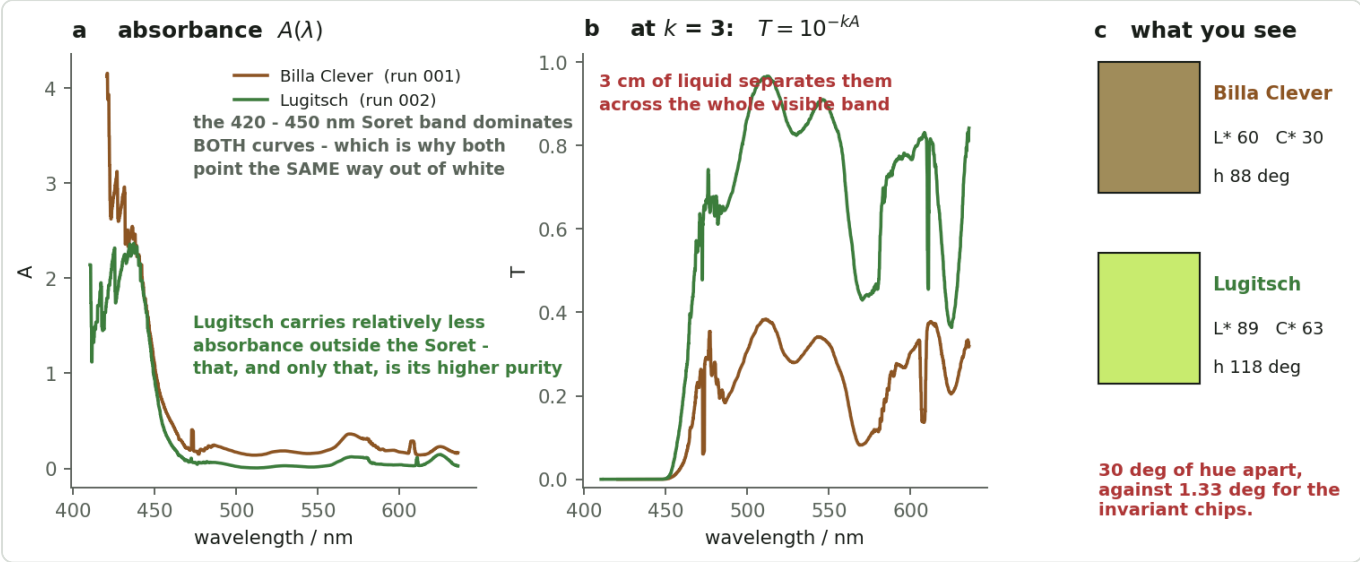


Figure 6 — $A(\lambda)$, the same curves at $k = 3$, and the resulting swatches.

	Billa Clever 001	Lugitsch 002	difference
L* lightness	59.8	89.2	29 points
C* chroma	30.0	62.8	33 points
h hue angle	88.1°	118.2°	30.1°
swatch	#a48e5b khaki	#caed72 yellow-green	

Thirty degrees of hue, against 1.33° for the invariant chips — a factor of twenty.

Two things produce that. First, keeping luminance restores the coordinate that was thrown away — *brownness is lightness is concentration*. Second, the exponential: a difference in absorbance *height* becomes a difference in transmitted *shape* once there is real liquid in the path.

⊖ This chip is deliberately NOT dilution-invariant, and it cannot be. A chip that shows "brownier" must move with concentration. The five chips above it stay invariant; this is the one you can hold up against a tube. Both are needed. Neither can be the other.

8. Beer–Lambert: what the law does and does not say

Section 7 applied $T = 10^{-(k \cdot A)}$ without saying where the law's authority ends. It matters, because A in this instrument is measured **against a reference**, and the formula never sees that reference.

8.1 The identity

Absorbance and transmittance are defined together:

$$A \equiv -\log_{10}(S/R) \quad \Rightarrow \quad 10^{-A} \equiv S/R$$

That is an **identity, not a physical claim**. Whatever **R** was, the **T** you recover is *"the fraction of the reference beam that got through"*. The law does not need to know the reference, because it returns an answer relative to that same reference. Nothing has been lost or assumed.

★ So the natural worry — *"the formula can't know what we referenced against, so mustn't the reference be standardised?"* — dissolves. **A** and **T** are two ways of writing one ratio, and both carry the reference with them.

8.2 Where the physics — and the convention — enter

The physics arrives one step later, when you assert

$$A(\lambda) = \varepsilon(\lambda) \cdot c \cdot \ell$$

★ **That** is where a convention is required, and it is not a standardised reference. It is the **blank**: identical vessel, identical solvent, identical geometry, everything except the analyte. Under a blank, **A** becomes the analyte's *own* absorbance, and $\varepsilon \cdot c \cdot \ell$ describes it with **c** the analyte's concentration.

Use a different blank and you get a different, equally valid **A** — it simply measures a different thing. The pinning is a **chemistry convention about how you prepare the reference**, not a property of the law.

The path multiplier stays consistent with this:

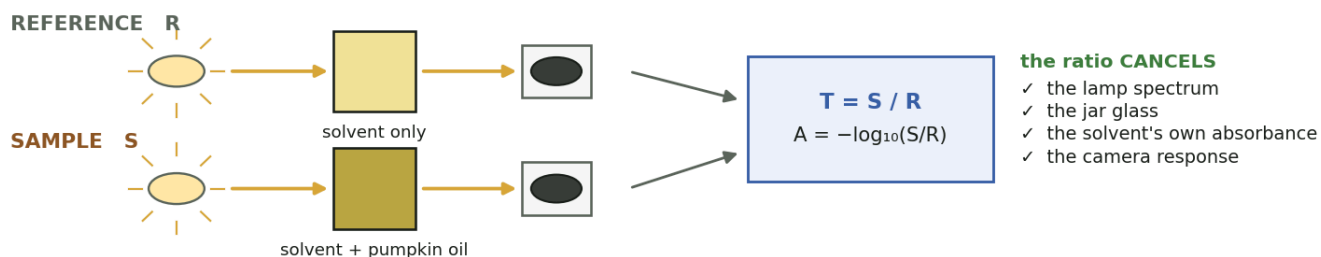
$$k \cdot A = \varepsilon \cdot c \cdot (k \cdot \ell)$$

⇒ multiplying **A** by **k** scales **the analyte only**, which is exactly right for a metric.

8.3 ★ For a colour chip, the reference is the illuminant

Rendering **T** under D65 says *"shine D65 through this filter"*. But **T** is relative to the blank, so what is really being drawn is **the reference beam, filtered by the extra pumpkin oil**. Change the blank and you change the light in the picture.

The reference is the illuminant



⇒ what the ratio leaves in the picture

the chip renders
 $D65 \times T^3$
 the pumpkin oil's EXCESS
 colour over the solvent
 Billa Clever, #a48e5b

the tube actually shows
 $D65 \times (\text{solvent})^3 \times T^3$
 never measured - it would need a
 blank-against-air capture

Runs 001 and 002 used SUNFLOWER OIL as the solvent, and sunflower oil is visibly yellow. Its colour divided out with everything else the ratio cancels. Against isopropanol - colourless in the visible - the two boxes above nearly coincide; against sunflower oil they do not. Both are legitimate answers to different questions, and one label is currently doing duty for both.

Figure 7 — what $T = S/R$ cancels, and what the ratio therefore leaves out of the rendered colour.

For an **isopropanol** blank this is harmless — isopropanol is essentially colourless in the visible, so "excess over the blank" and "what is in the tube" nearly coincide.

⊖ **Runs 001 and 002 did not use isopropanol.** They are the sunflower-oil fills (A_{valley} 0.1653 and 0.0181, matching `DOC_solvent_and_hue.md` exactly), and sunflower oil is visibly yellow. Its colour divided out with everything else the ratio cancels.

	shows
Transmitted from absorbance · ×3 path	the pumpkin oil's excess colour over sunflower oil
holding the tube up to the light	that, plus 3× the sunflower oil's own absorbance, plus the glass

Both are legitimate answers to different questions. Right now one chip is doing duty for both.

⚠ **And this makes §9.1's phone check less comfortable than it looked.** That photograph was of *these* fills, and the chip predicted it to 2.1° while having divided the solvent's colour out. One plausible reason, offered as a hypothesis and not a finding: **the phone's auto white balance does to the photograph roughly what the blank does to the measurement** — it removes a uniform colour cast. If so, the agreement is real but for a reason nobody designed, and it would not survive a fixed white-balance photograph with a grey card. Settling it needs one extra capture: **the solvent blank against air**, which yields the solvent's own A and can be added back.

8.4 ⊖ "3 cm" is a claim the vessel does not support

$k = 3$ triples the absorbance, which Beer–Lambert says is equivalent to tripling either concentration or path. Calling the result "3 cm" additionally asserts that the **measured** path was 1 cm.

It was not. The vessel is a **3 cm × 1.3 cm screw-jar with the beam running vertically**, so the optical path is the **fill depth** (`SPEC_capture_quality.md` §16.7.4):

poured	path	vs a full jar
4 ml	0.57 cm	44 %
6 ml	0.85 cm	65 %
8 ml — the standard fill	1.13 cm	87 %
9.2 ml — the brim	1.30 cm	full

At 8 ml, `k = 3` renders **3.4 cm**, not 3.0. And nothing in the protocol pins the volume: 1 ml $\approx$ 0.14 cm $\approx$ **11 % straight onto absorbance**.

⇒ What `path=3.0` honestly means is **"three times whatever was poured"** — which is why the chip is named `· ×3 path` rather than `· 3 cm`. Once §16.7.4's *fill to the brim and close the lid* pins the path at the jar's own 1.3 cm, `k` can be re-derived (2.3 for a true 3 cm) and the label can state a thickness.

△ Hue moves fast at that scale — §7.10 of `SPEC_color_retrieval.md` records -32.9° of rotation at 3 cm against -6.8° at 1 cm — so a 13 % path error is not obviously negligible.

9. How a chip is corrected — the five stages

Five stages, three of which exist to correct something specific. Stages 1, 3 and 4 are shared by every chip; stages 2 and 5 belong to the `×3 path` chip alone. △ Stage 3 was **not** always shared — §9.2 is the record of that, because it is the reason several numbers in this document changed.

#	stage	what it does	what it corrects
1	sanitize	non-finite $\rightarrow$ 0, negatives $\rightarrow$ 0, cap at the <i>relative ceiling</i> = $2 \times$ the spectrum's own 95th percentile	absorbance goes negative where $T > 1$, and negatives corrupt a CIE integral. The ceiling stops a $T \rightarrow 0$ spike in a few bins from dominating. It is <i>relative</i> so it survives a change of absorbance scale (gamma linearization moves it $\sim 2.2\times$) and a sealed-plugin version skew
2	path	<code>T = min(1, 10[^](-3·A))</code>	the declared viewing geometry — $3\times$ the poured depth (§8.4). The ceiling is applied <i>before</i> this: it exists to suppress a spike, not to cap thickness
3	dense resample	linear interpolation onto the 1 nm CIE grid; wavelengths outside 420–636 nm treated as transparent	★ the correction made on 2026-08-23 — see below
4	integrate	CIE 1931 2° observer under D65	—
5	read out	<code>XYZ $\rightarrow$ L*a*b* $\rightarrow$ L*C*h</code> for the numbers; <code>XYZ $\rightarrow$ sRGB</code> + clamp for the swatch	—

9.1 Stage 3, the one that mattered

The rig measures **420–636 nm**. Human vision runs to about **780 nm**. Something has to be assumed about the missing 40 % of the visible red, and the assumption **decides the answer**.

The colour library's default (`align`) fills gaps by **holding the boundary sample** — it takes the falling flank of the 624 nm band and extrapolates it flat across 636–780 nm, inventing that whole region from one point where the reference is only ~33 DN.

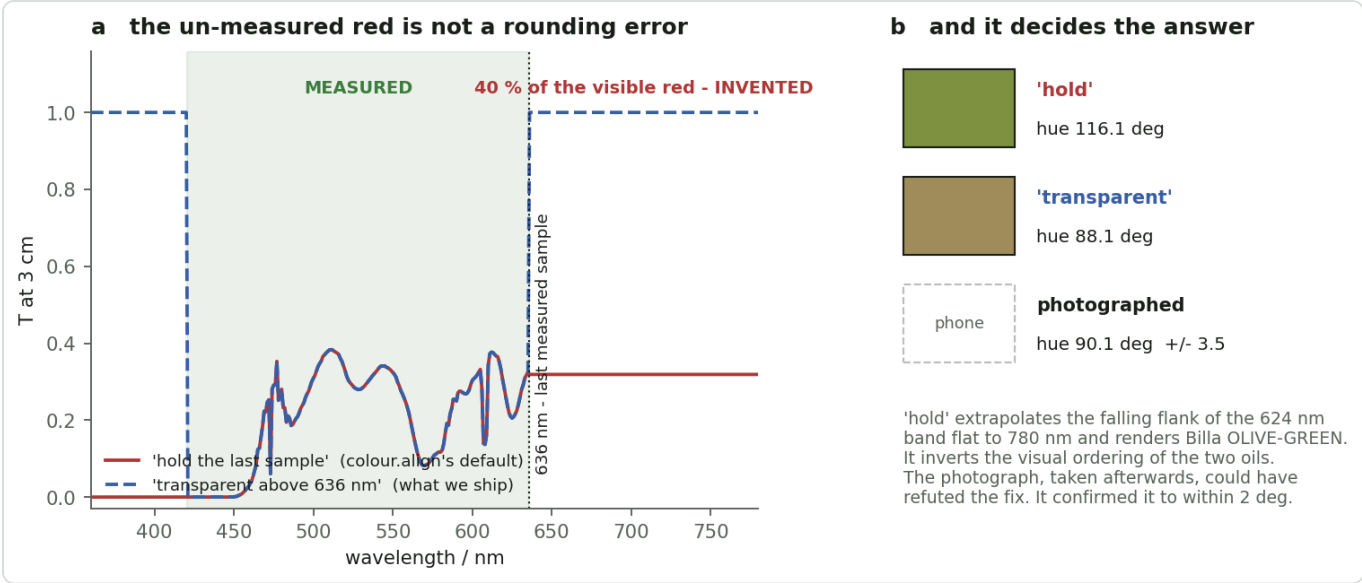


Figure 8 — the un-measured 636–780 nm, filled two ways, and the two answers that result.

red-tail assumption	Billa rendered at 3 cm	hue
<code>align</code> 's "hold the last sample"	#7f9341 olive-green	116.1°
"transparent above 636 nm" (shipped)	#a48e5b khaki brown	88.1°
photographed, uncalibrated phone	—	90.1° ± 3.5

➖ **"Hold" renders Billa Clever olive-green and inverts the visual ordering of the two oils.** The third option tried — sparse `T = 1` anchors plus the library's cubic interpolation — was worse still: it overshoots across the 1 nm step onto an opaque Soret and returns `L* = 144` , silently.

★ The photograph in §7.14 of `SPEC_color_retrieval.md` was taken **after** the fix was chosen, and could have refuted it. It confirmed Billa to within 2.1° and refuted "hold" by 26°.

⚠ **Still an assumption, and wrong at the blue end** — the Soret core below 420 nm absorbs *more*, not less. Under linear interpolation the CMF weight there is small, and transparency is the conservative choice in the red where the discrimination lives. This is a standing argument for extending the measured range.

✅ **9.2 For a year, stage 3 reached ONE chip out of eleven**

`__cieXyzDense` was called only from `spectrumToLab` . Every other chip went through `__cieXyz` , which called `align` — so **the constant-hold red tail stayed live on the ten invariant chips** for a year. Measured on the absorbed chromaticity, `align` 's hold against the explicit `A = 0` pad:

	legacy <code>align</code> -hold	<code>A = 0</code> padded	Δ hue	Δ C*
Absorbed Billa	h 310.86° · C* 165.9	h 308.54° · C* 144.5	2.31°	13 %
Absorbed Lugitsch	h 316.49° · C* 348.8	h 316.28° · C* 346.3	0.21°	0.7 %

⇒ Billa's absorbed chroma was **13 % inflated by invented red**, and λ_d moved almost 12 nm.

✓ **Fixed 2026-08-24 (P1).** `__cieXyz` now delegates to `__cieXyzDense`; `align` has no caller left in the colour path. Every number in §2, §2.1, §3.1 and §4 is post-P1.

➔ **One caller was deliberately NOT fixed.** The roast verdict runs through `spectrumToRgbAndHue` → `SpectrumToColorLogicModule`, which calls `align` itself. The constant-hold tail is still live there, on purpose: you do not move a shipped verdict on a colour argument. Recorded so it is not rediscovered as a bug.

⚠ **And P1 exposed a second padding trap the same day.** "Un-measured = transparent" is $A = 0$ for an absorbance but $T = 1$ for a transmittance. The first cut of `spectrumToChromaticity` padded everything the absorbance way, which rendered the un-measured red as **opaque** for the transmission chip and swung its θ_W by $\sim 13^\circ$. Caught by §4.2's own check — the 2.50° jumped to 16.58° — and fixed by making the source kind an explicit argument (`ABSORBANCE` / `TRANSMITTANCE`) rather than a default.

10. The family tree

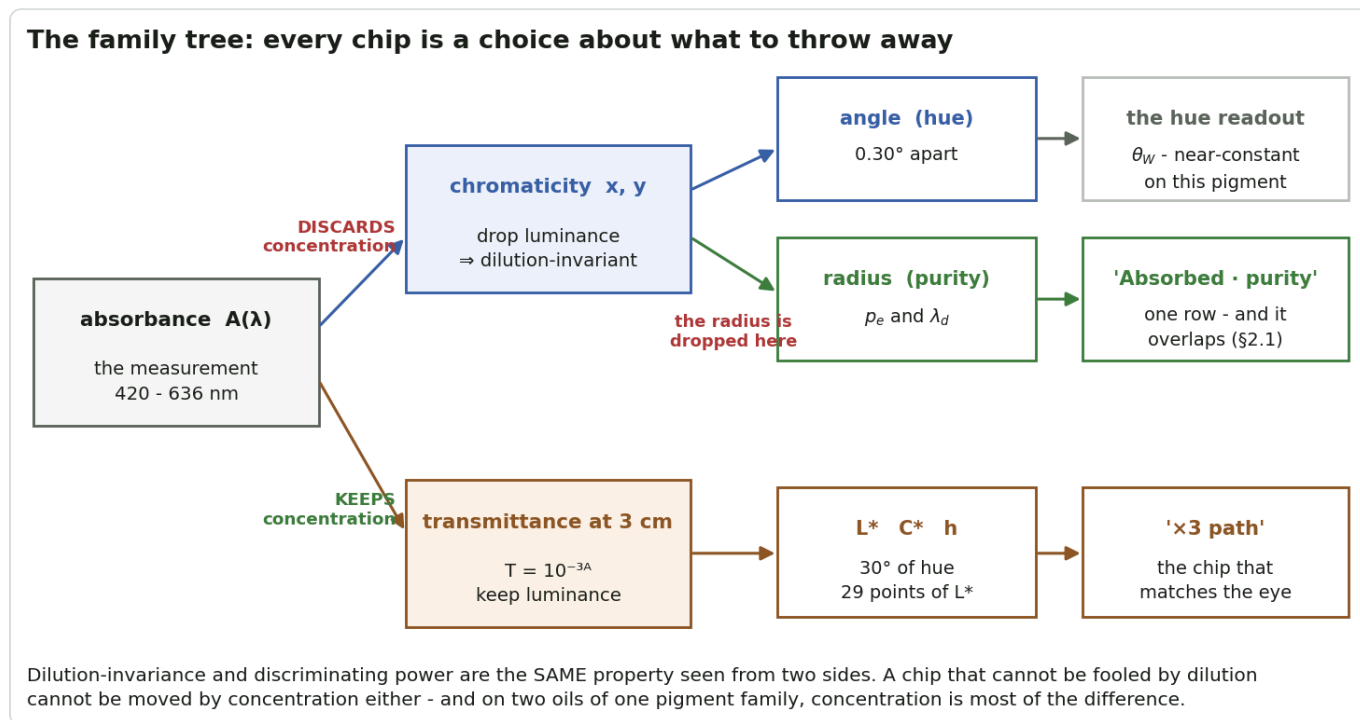


Figure 9 — every chip as one choice about what to discard.

Every chip in the set is one choice about what to discard. Read left to right:

- **Drop luminance** → invariant to dilution, and blind to it.
 - **Keep the angle** → θ_W : 1.33° of real signal here, and 5° across the whole archive (§3.1).
 - **Keep the radius** → `Absorbed · purity`: one row — and §2.1 shows it overlaps across the archive ($d = 0.55$), as does λ_d .
- **Keep luminance** → the `x3 path` chip: 30° of hue and 29 points of L^* , and it matches a photograph.

11. Why the chip set looks the way it does

Five design questions came out of §§1–10. Each is answered here by what the app now does, and by the measurement that settled it. The implementation record is §13.

11.1 It does not report a dominant wavelength

(λ_d, p_e) is the *honest* way to describe a chromaticity — no gamut, no dead axis — and for a while it looked like the invariant readout the chips were missing. It is not. The archive refutes both halves: p_e overlaps at $d = 0.56$, and λ_d is undefined on 31 % of the corpus, overlaps at $d = -0.08$ where it is defined, and correlates with the capture's blue edge at $r = 0.923$ (§2.1).

⇒ **The polar form describes; it does not discriminate.** Absorbed prints θ_w and p_e because they are the two coordinates of the thing being measured — not because either one tells you which oil is in the jar.

11.2 Chips whose colour cannot be shown are marked

Lab has no gamut, so the *numbers* are safe. The drawn swatch is still `XYZ → sRGB → clamp`, so on real oil it is a per-channel clamp of a colour no screen can produce — Lugitsch's Absorbed renders as pure magenta. Numbers and swatch disagreeing is worse than both being wrong together.

⇒ Every chip carries `MetricFieldViewStyle.isOutOfGamut`, drawn as a **dashed amber border** on screen and on paper. On real oil it is the normal case, not an edge case.

11.3 The processing rungs live in a sub-tab

Eight of the eleven original chips reported one quantity, and §3.1 measured that quantity as very nearly a constant of the pigment class — $\theta_w = 244.06 \pm 1.25^\circ$ across the whole archive. A rung that *moves* is still evidence the correction did something, so they are kept; they are shown under `Colour · processing rungs`, where they cannot be mistaken for the headline.

11.4 The names say what each chip is the colour OF

Intrinsic carries no meaning for a reader, and Perceived / As seen were near-synonyms for chips that mean opposite things (0.4° of separation against 30.1°).

was	is	family
Intrinsic	Absorbed	the light the oil removes
Intrinsic · purity	Absorbed · purity	
Intrinsic-perceived	Absorbed-complement	that colour reflected through white
Perceived	Transmitted-measured	read off the measured T
As seen · 3 cm	Transmitted from absorbance · ×3 path	Beer–Lambert at 3× the poured depth

Two rules hold throughout: the head is a hyphenated compound, and `·` introduces **exactly one** qualifier — which is why `measured` sits in the head rather than sharing the `·` slot with a rung. Absorbed also aligns the visible label with the vocabulary already in the code (`colorAbsorbed`).

 `· ×3 path` and not `· 3 cm`, because the beam is vertical and the path is the poured depth — see §8.4.

11.5 Every chip shares one CIE path

The 2026-08-23 red-tail fix reached one chip of eleven; the other ten held the boundary sample across 636–780 nm, inflating Billa's absorbed chroma by 13 % and moving its hue by 2.31° (§9.2). They now all go through `__cieXyzDense`.

11.6 ⚠ None of this makes the invariant chips separate the oils

Nothing can, without giving up the invariance — §3.1 is the reason, and it is chemistry rather than software. The separation is available: in `L*C*h` at a real viewing path, and in every metric in `SPEC_metric_research.md`. It is not available from a compass bearing.

12. ★ The colour model in use: Lab, with θ_w as the hue

HSL left the chip readout on 2026-08-24. Lab renders every swatch; θ_w is the reported hue on the chromaticity-only chips. This section is why, and what it costs.

chip	reported before (HSL)	reports now
Absorbed Billa	H 293.52° · S 100.0 · L 68.8	θ_w 244.4° · purity 66 %
Absorbed Lugitsch	H 300.00° · S 100.0 · L 50.0	θ_w 245.8° · purity 88 %
Absorbed-complement Billa	H 67.35° · S 100.0 · L 39.2	θ_w 64.4°
Absorbed-complement Lugitsch	H 68.77° · S 100.0 · L 39.6	θ_w 65.8°
Transmitted-measured Billa	H 70.85° · S 100.0 · L 81.6	θ_w 61.5° · purity 43 %
Transmitted-measured Lugitsch	H 70.55° · S 100.0 · L 82.2	θ_w 67.2° · purity 41 %
Transmitted from absorbance · ×3 path	already <code>L*C*h</code>	<code>L* 60 · C* 30 · h 88°</code>

⚠ Lab does not appear in the readout of the chromaticity-only chips, only in their swatch. That is the §12.3 result: h_{ab} is the worst of the three hue coordinates on the comparison these chips exist to make. "Lab everywhere" means Lab everywhere it renders.

What it buys. `S ≡ 100 %` is gone: purity separates the two oils 66 % against 88 % where `S` read 100.0 on both. Lugitsch's `H 300.00°` — the corner of the RGB cube, not a property of the oil — is gone with it. And the hue the chips report is now the chromaticity direction, which no gamut can clamp.

12.1 What it forces

-  The swatch still clamps. Lab has no gamut, so the numbers are safe — but the drawn patch is still `XYZ → sRGB → clamp`, and Lugitsch renders as pure magenta whatever the numbers say. Numbers and swatch disagreeing is worse than both being wrong together ⇒ every chip is marked (§11.2).
- ⚠ The achromatic guard had to be re-derived. `chroma(S, L) = (1 - |2L - 1|) · S < 8.0` has no meaning without HSL; it becomes a threshold on `C*`. `SPEC_color_retrieval.md` §7.13 recorded C2 as "NOT NEEDED — the HSL path is byte-for-byte unchanged". That premise is now void, and C2 reopens.
-  • hue-norm disappeared entirely. `__NORM_SATURATION` / `__NORM_LIGHTNESS` were HSL constants for re-rendering at a pinned saturation and lightness. The chips now render the chromaticity itself, so nothing replaces them.

-  **The verdict path is exempt.** `spectrumToRgbAndHue` — the `rgbxy` entry point — is called by `PumpkinOilPlugin.py:55`, and the roast verdict depends on its hue. It is a *different* method from the chip API and stays exactly as it is. "Retire HSL" means retire it from the **chip readout**.
-  **A tripwire was cut deliberately.** §7.13 asserted the `S = 100` defect in a test "so it cannot be fixed by accident without this section being revisited". Retiring HSL trips it on purpose — except that `spectrumToHsl` was KEPT for the sealed rows, so the test still passes, now guarding a method the chips no longer call. This section is the revisiting.

12.2 Order of operations — and it was the right order

The dense CIE path first, then Lab, because routing the chips through `__cieXyzDense` moves Billa's absorbed C^* from 165.9 to 144.5 (13 %) and its hue by 2.31° (§9.2). Every threshold the LCh switch needs — the achromatic cut, and every quoted purity — would otherwise have been derived against invented red and re-derived immediately.

 **It was done in that order (§13): P1 → re-derive → P3/P4.** Vindicated, and by more than expected: λ_d moved almost 12 nm, Billa's p_e from 68.7 % to 65.7 %, and h_{ab} 's penalty in §12.3 fell from 12.09° to 5.60° — a decision that would have been made on the wrong margin. **Every number in this document is post-P1;** the pre-P1 values survive only in §13.2's comparison column and in `tmp/oldPdfs/`.

12.3 Measured: on the invariant chips, LCh costs hue accuracy

 **This arrived after the decision and it argued against part of it — and it is what shipped.** §4.2 measured how well `Absorbed-complement` predicts the transmitted colour it stands in for, in each of the three candidate hue coordinates, over 88 archive runs:

hue coordinate	error	vs. shipping today
θ_w — chromaticity direction	$2.50 \pm 1.90^\circ$	1.7× better
H_{hsl} — the retired readout	$4.15 \pm 3.30^\circ$	—
h_{ab} — CIE Lab	$5.60 \pm 4.00^\circ$	 35 % worse

⇒ Switching the invariant chips to h_{ab} makes their hue number less accurate than the HSL it replaces, and it degrades exactly where it matters: where the complement is imaginary, h_{ab} goes to $13.48 \pm 1.31^\circ$ while θ_w improves to 1.68° .

The mechanism is the one §0 warns about. h_{ab} depends on chroma — that is what makes it perceptual — while θ_w does not. The complement sits at $C^*_{ab} \approx 100...200$; the transmitted colour it stands in for sits at ≈ 43 . Two colours can share a chromaticity direction *exactly* and still report different h_{ab} , purely from the chroma gap. **For comparing colours of very different chroma, Lab's hue angle is the wrong coordinate** — and that comparison is the entire job of this chip.

 **None of this touches the `x3 path` chip.** There L^* is a real measurement, C^*_{ab} is 30 and 63 (in range), and no cross-chroma comparison is being made. LCh is right there and stays.

The shape that keeps both properties: Lab everywhere for *presentation* — one model, no `S = 100`, no pure-magenta corner — with θ_w as the reported hue on the ten invariant chips. That is consistency without paying 3.1° for it.

⚠ **Caveat** — θ_w is the better **COORDINATE**, not a better **READOUT**. The 3.58° above is a *within-run* comparison (complement against transmitted), where any bias common to both cancels. As an absolute printed number θ_w is much weaker:

- it is **near-constant** — $244.06 \pm 1.25^\circ$ over the whole archive (§3.1), so the chip prints essentially the same figure on every oil anyone will ever measure;
- it carries the **same capture-edge confound as λ_d** — $r = -0.86$ with the first measured wavelength (against λ_d 's $+0.923$);
- it does **not** separate the classes — $d = -0.02$, overlapping.

★ Its one structural advantage over λ_d is that it is **always defined**: 88 of 88 runs, against 61 of 88, because a ray that exits through the purple line still has an angle.

⇒ Choose θ_w because it represents the chromaticity relationship most faithfully and never goes undefined — **not** because the printed number will tell anyone which oil is in the jar. Nothing in the invariant family does that.

13. ★★ AS BUILT — 2026-08-24

All phases implemented. 583 tests green (520 app + 63 plugins), **SDK_VERSION** unchanged at 1. All 203 archived reports regenerated onto the new chips (§13.3).

phase	what shipped
P0	The sealed <code>DevSpectralPlugin 1.1.0</code> row deleted from <code>db_plugin</code> . The registry falls back to the built-in, so the bench runs the working tree. The two <code>PumpkinOilPlugin</code> rows are untouched.
P1	<code>__cieXyz</code> delegates to <code>__cieXyzDense</code> . <code>align</code> has no caller left in the colour path (§9.2).  Changed every chromaticity number in this document.
P2	Sweep and figures re-derived against P1; the sweep keeps an explicit <code>legacyAlignXy</code> so the comparison cannot silently collapse to one column.
P3	Core gained <code>spectrumToChromaticity</code> , <code>complementOf</code> , <code>directionFromWhite</code> , <code>purityOf</code> , <code>lchOf</code> , <code>isRenderable</code> , <code>ACHROMATIC_CHROMA_LAB = 7</code> . ★ Purely additive — <code>spectrumToHsl</code> / <code>rgbFromHsl</code> / <code>chroma</code> all stay, because three sealed rows call that surface through a strict-equality SDK gate.
P4	The dev plugin reads Lab for presentation and reports θ_w as hue. No <code>spectrumToHsl</code> call remains in the plugin; <code>__NORM_SATURATION</code> / <code>__NORM_LIGHTNESS</code> are gone.
P5	<code>MetricFieldViewStyle.isOutOfGamut</code> , rendered as a dashed amber swatch border in both renderers, so paper carries the same warning as the screen.
P6	 Dropped — §2.1 refuted λ_d .
P7	Renamed and reordered; the eight hue-norm rungs moved into a <code>Colour · processing rungs</code> sub-tab.
P8	This section, <code>SPEC_color_retrieval.md</code> §7.15, and the whole-archive re-render (§13.3).

13.1 The chip set as it now reads

chip	prints
Transmitted from absorbance · ×3 path	L* 60 · C* 30 · h 88° — the full LCh triple, every number a measurement
Absorbed	θw 244.4° · purity 66 %
Absorbed-complement	θw 64.4° — hue only (§4.1)
Transmitted-measured	θw 61.5° · purity 43 %
Absorbed · purity	66 % — demoted, kept as the free predictor for the complement
sub-tab Colour · processing rungs	the de-spiked and baseline recomputations of the two absorbance chips

(Billa Clever, run 001. Lugitsch reads L* 89 · C* 63 · h 118° / θw 245.8° · purity 88 %.)

13.2 What P1 moved, and what it did not

	before P1	after P1
Billa θ _w	245.67°	244.44°
Billa p _e	68.7 %	65.7 %
Billa λ _d	433 nm	444 nm
the two oils' θ _w gap	0.30°	1.33°
complement vs transmitted (§4.2)	3.58°	2.50° — better
complements outside the locus	13 of 88	9 of 88
×3 path chip	L* 60 · C* 30 · h 88°	unchanged

★ The ×3 path chip not moving is the check that P1 did what it claimed: that chip was already on the dense path, so it *must* be byte-identical. It is.

⚠ **Every conclusion survived P1; several margins shrank.** λ_d is still refuted, purity still overlaps, θ_w is still the best hue coordinate and h_{ab} still the worst — but h_{ab}'s penalty fell from 12.09° to 5.60°, so §12.3's decision is right on a narrower margin than it first looked.

★ **13.3 The whole archive regenerated — and the regression it caught**

All 203 archived measurement reports re-rendered onto the new chip set, 0 failures
(diagnostics/regenerate_reports.py --write), each rebuilt from its own embedded workflow.json and re-evaluated by today's plugin. The spectra are the originals — nothing was re-measured. The pre-2026-08-24 files are at spectracs-references/tmp/oldPdfs/, an exact 205-file mirror.

🚫 **oldPdfs lives INSIDE the archive root**, which every diagnostic walks. Left alone, each run would have been counted **twice** and every archive statistic silently corrupted — including the 88-run sweep this whole document rests on. EXCLUDED_DIRS = {"oldPdfs", "discussion"} is now honoured by peak_ratio_archive, all_metrics_archive, regenerate_reports, report_reconstruct and settling_sweep, and reportPaths() was re-checked at 203 after the copy existed. ⚠ The earlier backups (tmp_backup_*) sit OUTSIDE tmp/ for exactly this reason.

★★ The bulk run caught a regression 582 unit tests did not. Two runs of the 20260806A null series carry an absorbance that is negative at every wavelength, so it sanitizes to nothing. §7.10's F10 and F13 are different cases, and the rewrite had collapsed them:

spectrum	before the rewrite	after (wrong)	now
missing	no row	no row	no row
present, no positive signal	grey achromatic / undefined	🚫 row vanished	grey achromatic / undefined

⇒ A null run looked like a run whose colour had simply *not been computed*. Fixed, and asserted by `test_a_spectrum_with_no_positive_signal_still_renders_a_chip`.

⚠ One deliberate change beyond the phase list. The `×3 path chip` returned nothing on such a sample too — pre-existing, and tolerable while it sat *last*. It now **leads** the list, so a silent gap at the top reads as "not computed" rather than "there is no colour here". It now renders the same grey row as every other chip. Recorded here and in `SPEC_color_retrieval.md` §7.15.5 so it is a decision rather than a drift.

⇒ ★ Two of the four checks in this document came from running the code over the whole archive rather than from reasoning about it — §2.1's refutation of λ_d , and this regression. The unit suite is green in both cases.

13.4 Owed

- II Rig click-through. Nothing here has been seen on the bench; the evidence is 583 tests and the as-built numbers above.
- ✓ The PDF report path is verified. The chips render in the new order through `MatplotlibWorkflowRenderer`, and the out-of-gamut marker draws as a dashed amber border on paper — present on `Absorbed` and `Absorbed-complement`, absent on the `×3 path chip`, whose swatch is honest.
- ⚠ Old saved runs keep their old labels — the label is data inside the `DbMeasurement` blob and there is no migration. A reloaded pre-2026-08-24 run shows `Intrinsic`, `As seen · 3 cm` and the rest. ★ The archived *reports* do not have this problem: they were rebuilt from their spectra.
- ⚠ The android trees (`android/server/app_src`, `android/spike/app_src`) carry their own stale copies of `EvaluationColorUtil` and were deliberately not touched.

14. Glossary

term	meaning
chromaticity (x, y)	colour with brightness divided out — a point on the horseshoe map
spectrum locus	the curved edge of the horseshoe: the pure single wavelengths
white point	the chromaticity called "white"; we use D65, (0.313, 0.329)
dominant wavelength	the wavelength where the ray from white through a colour leaves the locus
excitation purity	the fraction of the way from white to the locus along that ray, in percent
gamut	the set of colours a given device can actually produce
complement	the chromaticity that mixes with a colour to give white

term	meaning
Soret band	the very strong 420–450 nm absorption of chlorophyll-type pigments
L*a*b* / L*C*h	the perceptually spaced CIE colour space, in Cartesian / polar form
CMF	colour matching functions: the standard observer's three cone responses
transmittance T	the fraction of the reference beam that gets through: $T = S/R$
absorbance A	$-\log_{10}(T)$; additive, and linear in concentration $\times$ path
blank	the reference preparation: same vessel, same solvent, same geometry, no analyte
Beer–Lambert	$A = \epsilon \cdot c \cdot l$ — absorbance is proportional to concentration $\times$ path length
ϵ	molar absorptivity: how strongly the analyte absorbs at each wavelength

15. Provenance

Every number in this note is computed by the shipped `EvaluationColorUtil`, never re-derived. Two corpora feed it, and which one a claim rests on is what decides how much weight it carries:

- **the two runs** of §2 and §5–§9 — read back from their own report PDFs, through the embedded `workflow.json`, i.e. the shipped artefact;
- **the 88 labelled isopropanol archive runs** of §2.1, §3.1 and §4 — the same corpus `SPEC_metric_research.md` §12 scores, with its labelling, exclusions and diffuser split unchanged.

★ Everything in this note that *survived* did so against the second. Everything refuted — the purity gap, λ_d , the complement's renderability — was proposed on the first.

The one deliberate exception is the "hold" red tail in §9.1, which exists to be refuted.

```

runs    spectracs-references/tmp/20260823_newchips/
        001_BillaClever_newchips.pdf, 002_Lugitsch_newchips.pdf
        ^ REGENERATED 2026-08-24 on the post-P1 plugin: their embedded
        workflow.json now carries the four-chip set, so every number in
        this document reproduces from them directly.
        pre_P1_2026-08-24/    the originals, kept -- the only place the
        "before P1" column of §13.2 can still be reproduced from.
archive spectracs-references/tmp/                203 reports, 88 labelled
        ^ ALL REGENERATED 2026-08-24 on the post-P1 plugin (§13.3)
        -> dominant_wavelength_archive.csv
        spectracs-references/tmp/oldPdfs/        the 205-file pre-P1 mirror
        ^ EXCLUDED from every archive walker -- it sits inside the root
code    spectracsPy-core    ../plugin_sdk/util/EvaluationColorUtil.py
        spectracs-plugins  ../plugins/dev/DevSpectralPlugin.py
                __colourChips, __asSeenChip
scripts diagnostics/dominant_wavelength_archive.py    the archive sweep
        diagnostics/peak_ratio_archive.py            its loader and labelling
        diagnostics/regenerate_reports.py            the archive re-render
        diagnostics/colour_geometry_figures.py        -> docs/figures/colour_*.png
spec    docs/SPEC_capture_quality.md
        §16.7.4 the fill level IS the path length
        docs/SPEC_color_retrieval.md
        §7      the pipeline and why it discards the difference
        §7.12   the C1-C7 phases
        §7.13   as built, 2026-08-23
        §7.14   the phone check
        §7.15   as built, 2026-08-24 -- the contract for §§11-13
        §7.16   Appendix A as specification, with E1/E2 as owed work

```

★ Appendix A — Why the SOLVENT changes what the EYE can see

(2026-08-24 · **argument, not measurement** · *SPEC_color_retrieval.md* §7.16 carries the same content as the specification, with the experiments as owed work)

Two observations, both Edwin's, both from the bench rather than the archive.

1. After many eprouvettes of pumpkin oil in isopropanol, switching the solvent to sunflower oil made the difference between the green and the brown oil obvious **by eye, immediately**.
2. **The red peak became more pronounced** in sunflower oil — and the same had been seen in the **white spirit** session.

△ The second observation is the more diagnostic of the two, and it is the one that identifies the mechanism: it happened in *both* solvents. White spirit ($n \approx 1.44$) and sunflower oil ($n = 1.473$) have nothing in common except that both are **nonpolar and index-matched to the oil**, so both DISSOLVE it where isopropanol only emulsifies it.

That appears to contradict everything in §§2.1, 3.1 and 11 — which measured, over 88 runs, that colour does *not* separate these oils. It does not contradict it. §A.1 is why.

➡ **Nothing in this appendix is measured.** It is a mechanism, offered because the archive's silence needs explaining rather than waving away. §A.5 is what would settle it.

A.1 ★★ The instrument cannot see what the eye sees — by construction

$T = S/R$, and R is the **solvent blank** (§8.3). Two consequences, both design rather than defect:

- 1. **The solvent's own colour divides out exactly.** Sunflower oil is visibly yellow; the measurement removes it. Every colour number in this document describes *the pumpkin oil's excess over the solvent*. The eye in front of an eprouvette sees **the whole liquid**.
- 2. **A scattering veil largely divides out too**, because the blank is the same preparation carrying the same turbidity.

⇒ **The archive's silence is not evidence against the observation — it is a different quantity.** The two mechanisms below are invisible to $T = S/R$ on principle. This is the same point §8.3 makes about the reference being the illuminant, seen from the other side.

A.2 The dominant mechanism: refractive-index matching

	n	$m = n_{oil}/n_{medium}$	scattering factor
pumpkin oil	1.47	—	—
isopropanol	1.377	1.0675	1.98×10^{-3}
sunflower oil	1.473	0.9980	1.85×10^{-6}

where the factor is

$$\left(\frac{m^2-1}{m^2+2} \right)^2$$

⇒ **about 1070× less scattering in sunflower oil.** In isopropanol the oil never dissolves — it emulsifies, and every droplet is a lens. Like into like, the droplets become optically invisible even where they are physically present.

A.3 ★ Why scattering destroys a colour DIFFERENCE, not just brightness

Scattering puts a floor under the transmittance, and the floor kills the exponential.

$$T_{obs}(\lambda) = T_{direct}(\lambda) + S$$

S is forward-scattered light. It took short, randomised paths through little absorbing material, so it is **much whiter than the direct beam**.

In a clear liquid at 3 cm, $T = 10^{-kA}$ ranges over orders of magnitude — the deep bands go genuinely black and the windows stay bright. That dynamic range **is** the saturation, and it is the entire mechanism §7 rests

on. With a veil, T can never fall below S : the deep parts of the spectrum are clamped, the spectrum flattens, and the chromaticity walks toward white.

⇒ Both samples become pale yellow-grey, and pale colours are hard to separate without a side-by-side reference. **That is a sufficient explanation of observation 1.**

➡ It is NOT an explanation of observation 2, and an earlier draft of this appendix claimed it was. The contrast reduction is by the factor

$$\frac{T_{\text{base}}}{T_{\text{base}} + S}$$

which bites hardest where T is *lowest*. That is the Soret, at $T \approx 0.03$ — not the red bands, which sit at $T \approx 0.6$ and would barely notice a veil. The floor predicts scattering should flatten the **blue** and spare the **red**; observation 2 is the opposite. §A.5 is the mechanism that predicts the right asymmetry.

A.4 The second mechanism, which also favours sunflower

Transmittances multiply: $T_{\text{total}} = T_{\text{solvent}} \times T_{\text{pumpkin}}$.

Sunflower oil is yellow — it absorbs blue. And **blue is where the two pumpkin oils agree**: the Soret dominates both, which is why θ_w is near-constant across the archive (§3.1). In the red, sunflower is transparent, so the pumpkin difference passes undiminished.

⇒ The solvent acts as a **filter that suppresses the band where the oils are identical and passes the band where they differ**. Isopropanol, being colourless, leaves the uninformative blue residue in the light reaching the eye, diluting the contrast.

➡➡ A.5 Observation 2 — a mechanism proposed, tested, and refuted the same day

⚠ **The phenomenon was already established**, on 110 fills, in `SPEC_metric_research.md` §12.6: on Soret-normalised absorbance the 568 nm band runs 0.087–0.213 across 106 isopropanol fills and 0.235–0.289 across four white-spirit fills — no overlap — with turbidity already ruled out ($r = -0.016$) and the cause recorded as **unsettled**. What follows removes one candidate; it does not settle it.

Proposed: a grating maps input angle onto wavelength, so an emulsion's ~17° forward lobe broadens the effective linewidth — narrow features wash out, broad ones survive. It has the right shape.

Tested. ★ **Convolution conserves area.** Blurring lowers a band's height and widens it, leaving the integral alone. So measure the 624 nm band's **area** above its local chord as a fraction of the **Soret** area — dose-free, since both scale with concentration:

	n	A_Soret (dose check)	624 area / Soret area
index-matched (white spirit + sunflower)	7	0.877 ± 0.263	0.0221 ± 0.0107
isopropanol	72	0.762 ± 0.146	0.0013 ± 0.0010

⇒ **16.6× at matched dose** ($d = +6.65$, ranges separate; the two A_Soret populations differ by 1.2×).

➡ A 16.6× dose-free AREA difference cannot be convolution. Refuted by its own prediction.

➡ And every other optical candidate fails for one shared reason — they all hurt the **SORET** more, so each predicts a *larger* Soret-normalised Q band in the emulsion, not a 16.6× smaller one: veiling glare scales

contrast by $T_{\text{base}}/(T_{\text{base}}+S)$, severe at the Soret and mild at 624 nm; the package/sieve effect flattens the strongest bands most, which §12.6 had already noted.

⇒ **Nothing optical survives.** At the same pigment load the 624 nm transition carries 16.6× less area relative to the Soret in isopropanol — a change in the pigment's own state rather than in how the light reaches the detector. II Which is exactly where §12.6 left it.

△ A.5.1 What the test could not do

The first cut measured **equivalent width** $W = \text{area}/\text{height}$ and reported isopropanol's band as 3× narrower. — **Confounded — do not quote it.** Within isopropanol $r(W, \text{height}) = +0.543$, and W rises across height tertiles (1.63 → 2.21 → 2.70 nm): a band near the noise floor has its peak set by a noise excursion. And the height populations **do not overlap at all** — every isopropanol band is fainter than the weakest index-matched one — so there is no fair slice to compare on.

△ **The diffuser A/B could not validate W either.** It is the archive's one known blurring event, so W should have risen — but the diffuser erases the band completely (no band above the chord on all five diffuser-IN runs), leaving W undefined. **W has never been shown to detect blurring on this instrument.** The conclusion above rests on AREA, which needs no such demonstration.

A.6 △ Where to hold back, and the experiments

△ **All three arguments push the same way, which means none of them is tested by the observation.** And the observer's own caveat is the right one: a side-by-side pair in one light is far more sensitive than sequential viewing on different days. The eye adapts, and memory for colour is poor.

E1 — the premise, ten seconds. Shine a narrow beam sideways through each eprouvette in a dark room. The isopropanol emulsion will show a visible **Tyndall beam**; the sunflower solution will show nothing. Tests "is it scattering?" directly, with no colorimetry at all.

E2 — the claim. Four eprouvettes at once, same light, same day: green and brown oil × isopropanol and sunflower, **matched pigment dose**. Judge the *pairs*, not the tubes. If the sunflower pair separates and the isopropanol pair does not at equal dose, the mechanism is confirmed and the archive's silence is explained rather than contradicted.

△ **Match the dose carefully** — sunflower oil is far more viscous, so a capillary delivers a different mass into each, and concentration would confound the whole comparison.

— **E3 as first written is unrunnable, and has been replaced.** It said to measure the WIDTH of a narrow lamp line under two solvents. **There is no such line in this beam** — both candidates are detector artefacts. 608–610 nm is a Bayer channel crossover; the 473 nm "blue-pump edge" rises in **one or two samples** (0.15–0.29 nm against a 0.146 nm grid), far below any plausible instrument linewidth, and sits at **472.5 nm in the REFERENCE and 473.1 nm in the SAMPLE of the same run.** An optical feature cannot move; a threshold where the dominant Bayer channel switches moves with signal level.

E3 — ★ the split sample. §A.5 leaves a chemistry hypothesis with every optical explanation removed, and §12.6 leaves the cause unsettled. Both are limited by the same confound: 7 index-matched fills of one oil pair against 72 isopropanol runs spanning a year, two rigs and a rebuild.

⇒ **One oil, one dilution, split, half into each solvent, same session, same rig.** Report $\text{area}(624)/\text{area}(\text{Soret})$ — dose-free, so the split need not be perfect. Two spectra, one evening, and every confound above disappears at once. △ Use the **green oil**: its 624 band is the taller, so the isopropanol arm sits furthest from the noise floor where §A.5.1's problems live.

⇒ ★ **If E2 and E3 confirm it, the consequence for this document is specific and limited:** none of §§2.1, 3.1 or 11 changes, because they measure the excess over the blank. What changes is the reading of §7 — the `×3 path` chip would be understood as showing *less* than the eye does, by the whole of the solvent's contribution, and §8.3's "the reference is the illuminant" becomes the most practically important paragraph in the document rather than a caveat.